bearing on the physics of string theory. In fact, based on the discussion in Chapter 10 in which we found that circular dimensions have a minimum radius, one might be tempted to say that string theory does not allow the sphere in Figure 11.3 to shrink all the way down to a pinched point. But remember, as also noted in Chapter 10, that if a chunk of space collapses—in this case, a spherical piece of a Calabi-Yau shape—as opposed to the collapse of a complete spatial dimension, the argument identifying small and large radii is not directly applicable. Nevertheless, even though this idea for ruling out flop transitions does not stand up to scrutiny, the possibility that the fabric of space could tear still seemed rather unlikely.

But then, in 1991 the Norwegian physicist Andy Lütken together with Paul Aspinwall, a graduate-school classmate of mine from Oxford and now a professor at Duke University, asked themselves what proved to be a very interesting question: If the spatial fabric of the Calabi-Yau portion of our universe were to undergo a space-tearing flop transition, what would it look like from the perspective of the mirror Calabi-Yau space? To understand the motivation for this question, you must recall that the physics emerging from either member of a mirror pair of Calabi-Yau shapes (if selected for the extra dimensions) is identical, but the complexity of the mathematics that a physicist must employ to extract the physics can differ significantly between the two. Aspinwall and Lütken speculated that the mathematically complicated flop transition of Figures 11.3 and 11.4 might have a far simpler mirror description—one that might give a more transparent view on the associated physics.

At the time of their work, mirror symmetry was not understood at the depth required to answer the question they posed. However, Aspinwall and Lütken noted that there did not seem to be anything in the mirror description that would indicate a disastrous physical consequence associated with the spatial tears of flop transitions. Around the same time, the work Plesser and I had done in finding mirror pairs of Calabi-Yau shapes (see Chapter 10) unexpectedly led us to think about flop transitions as well. It is a well-known mathematical fact that gluing various points together as in Figure 10.4—the procedure we had used to construct mirror pairs—leads to geometrical situations that are identical to the pinch and puncture in